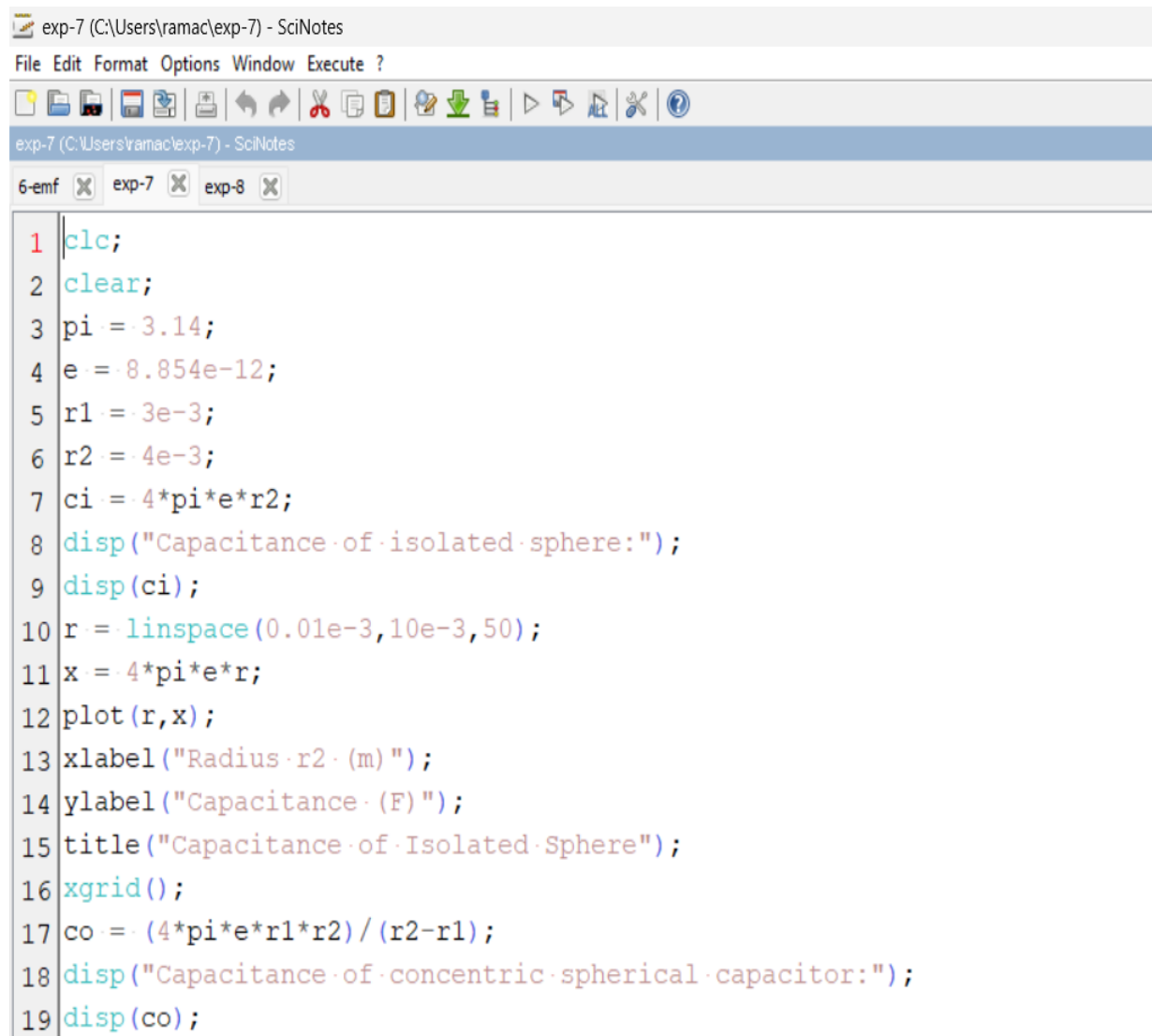


Exp:7-Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

Aim: To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

Software required: SCILAB version 6.0.1 **Practical**

output:



```
exp-7 (C:\Users\ramac\exp-7) - SciNotes
File Edit Format Options Window Execute ?
exp-7 (C:\Users\ramac\exp-7) - SciNotes
6-emf exp-7 exp-8

1 clc;
2 clear;
3 pi = 3.14;
4 e = 8.854e-12;
5 r1 = 3e-3;
6 r2 = 4e-3;
7 ci = 4*pi*e*r2;
8 disp("Capacitance of isolated sphere:");
9 disp(ci);
10 r = linspace(0.01e-3, 10e-3, 50);
11 x = 4*pi*e*r;
12 plot(r, x);
13 xlabel("Radius r2 (m)");
14 ylabel("Capacitance (F)");
15 title("Capacitance of Isolated Sphere");
16 xgrid();
17 co = (4*pi*e*r1*r2) / (r2-r1);
18 disp("Capacitance of concentric spherical capacitor:");
19 disp(co);
```

Scilab 6.1.1 Console

File Edit Control Applications ?



Scilab 6.1.1 Console

"Capacitance of isolated sphere:"

4.448D-13

"Capacitance of concentric spherical capacitor:"

1.334D-12

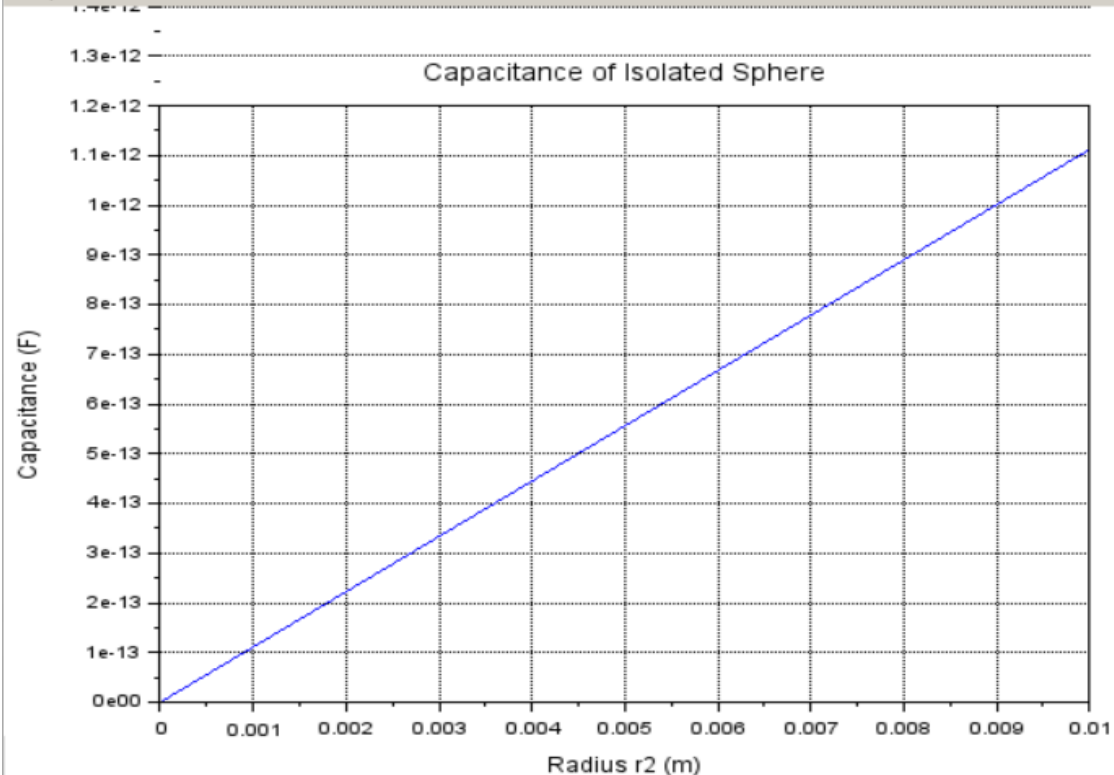
-->

Graphic window number 0

File Tools Edit ?



Graphic window number 0



Theoretical Output:

⑦

$$r_1 = 3 \times 10^{-3}$$
$$r_2 = 4 \times 10^{-3}$$
$$\epsilon_0 = 8.854 \times 10^{-12} \text{ f/m}$$
$$\pi = 3.14$$

1) formula $\Rightarrow C = 4\pi\epsilon_0 r^2$

$$\Rightarrow 4 \times 3.14 \times 8.854 \times 10^{-12} \times 4 \times 10^{-6}$$
$$\Rightarrow 4.448 \times 10^{-3}$$

2) $C_0 = \frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1}$

$$\Rightarrow \frac{4(3.14)8.854 \times 10^{-12} \times 3 \times 4}{4 \times 10^{-3} - 3 \times 10^{-3}}$$
$$\Rightarrow 1.334 \times 10^{-12} \text{ F}$$

$r_2 - r_1$
 $\Rightarrow 1 \times 10^{-3} \text{ m}$

Result :

Thus the program was verified for Capacitance of an isolated sphere and two concentric sphere using SCILAB was successfully.